Scheduler + phone FFN: progress report

2026-09-25 17:14:10 UTC | RTX 4060 Ti + i9-12900K + OP15 + Pixel 10 Pro

58.7% lower host energy on today's trace

Two-phone completion PASS, 94.414 kJ in 1833.0 s; 58.69% host saving. Exact output check FAIL: 13/14. Baseline: 228.535 kJ / 2,244.348 s.

26.7% less host energy on the longer trace

Dispatcher-only: 536.32 -> 393.04 kJ; 4,984 -> 3,381 s. 31/31 requests complete; 14/31 exact output sequences. One matched pair.

Short-trace phone savings exceed 25%, with substantial variation

OP15: 34.6-46.1% (mean 40.3%). OP15 + Pixel: 24.4-44.3% (mean 34.3%). Same desktop + dispatcher reference: 63.97 kJ. Only 4-5/9 exact sequences.

New OP15-only control: positive two-phone energy difference

OP15 104.46 -> two phones 94.41 kJ (9.6% less); 0.6% longer duration. One run each. Exact outputs vs legacy: 12/14 and 13/14.

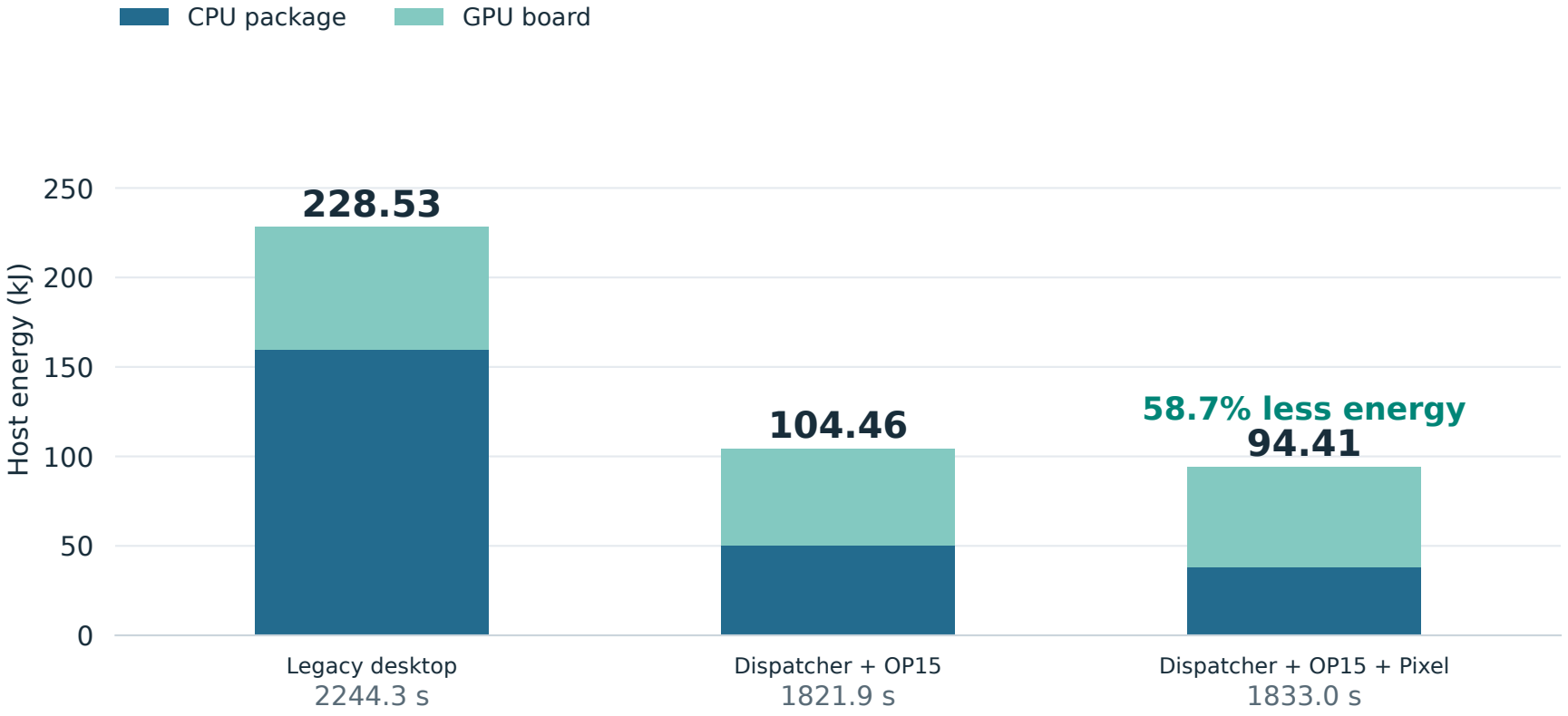
Pixel layer round trips are 2.72-3.04x faster

B1/B2/B4: 38.42/61.44/88.39 -> 13.56/20.20/32.45 ms. Kernel outputs unchanged. Separate server check: 17.4% host saving, 4/4 token-identical outputs.

Host energy is measured CPU-package RAPL plus GPU-board NVML over the paid trace interval. It is not wall-plug energy and excludes phone energy. Phone power is only modeled (4.5 W active, 0.875 W idle), so no measured whole-system saving is claimed. Compare percentages only within a trace and its stated baseline. The two longtail desktop arms and the dev_v2 baseline are single measurements; there are two repeats of each dev_v2 phone configuration. No confidence interval is established.

Today's completed two-phone trace

longtail_eval_v2: 14 requests / 3,604 output tokens | Measured host energy

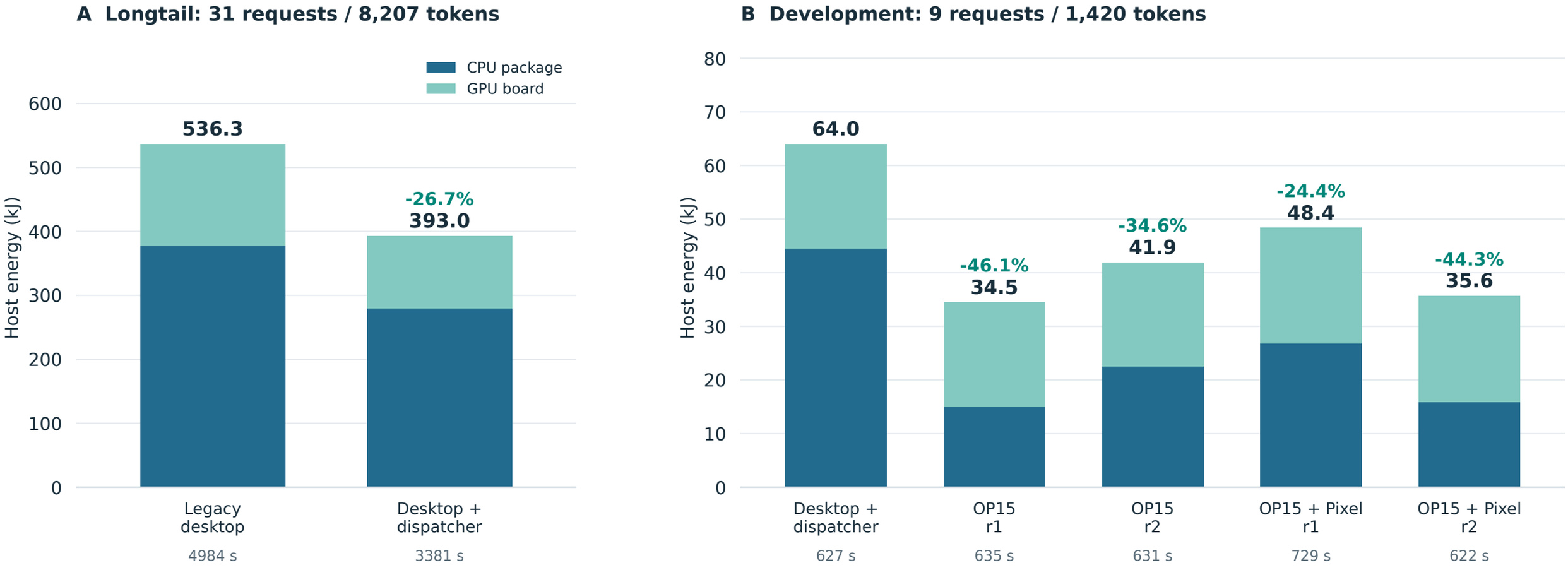


Completion PASS 14/14; strict output identity FAIL 13/14.

Single run per configuration. Phone energy excluded. Two phones use 9.6% less than OP15; OP15 exact outputs 12/14; two-phone exact outputs 13/14. Both strict checks FAIL.

Measured host energy: compare within each trace

CPU package (RAPL) + GPU board (NVML) | Lower is better | Phones excluded

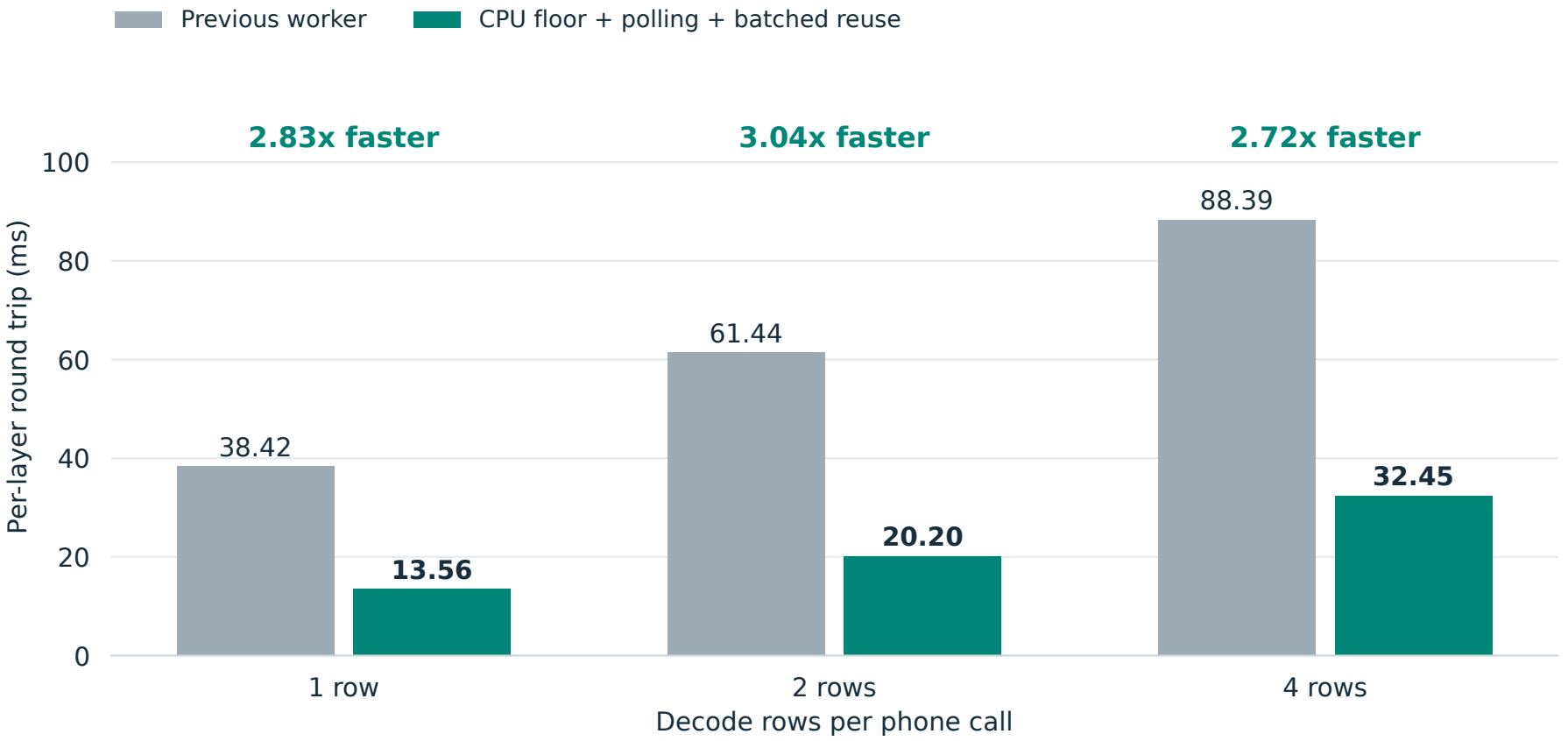


A: completion 31/31; exact outputs 14/31. B: completion 9/9; exact outputs 4-5/9.

One longtail pair; two phone repeats on dev_v2. No confidence intervals. Different traces and baselines; savings do not add.

Pixel FFN: fix the cost of short CPU bursts

Server-like cadence, real Qwen layers 18-23, packed CPU weights, ADB TCP



Before: mean of two control medians. After: optimized median; 72 calls per row count.
Byte-identical kernel outputs. These are layer-call latencies, not whole-model token latencies.